

ME 516: Polymer Technology for Engineers

Course Code	ME 516
Course Name	Polymer Technology for Engineers
L-T-P-C	3-0-0-3
Prerequisites	Materials Science for Engineers (IC241) or equivalent material science course
Intended for	UG/PG
Mutual Exclusion	None
Approval	recommended by the 39th BoA (Mar 2021)

Course Contents

Introduction to Polymer: Brief history of polymers; polymer classification; molar mass and molar mass distribution; polyethylene-polymerization process; chemical bonding- primary, secondary and tertiary structure; crystallinity. (3 Lectures)

Thermal Transitions in Polymers: Glass transition temperature; factor affecting glass transition temperature- molar mass dependence, plasticization, chain entanglement effect; crystalline melting point; differential scanning calorimetry. (4 Lectures)

Mechanical Properties of Polymers: Stress-strain measurements; dynamic modulus; methods of measurements of polymers- tensile testing, fracture, creep, relaxation time, compression; effect of structural and environmental factors on mechanical properties- molecular weight, cross-linking, crystallinity, strain rate, temperature. (6 Lectures)

Viscoelastic Properties of Polymer: Simple rheological responses- ideal elastic, pure viscous and rubberlike elastic; viscoelasticity; mechanical models for viscoelastic responses- Maxwell model, Kelvin-Voigt model, Four-parameter model; relaxation time; material response time-Deborah number. (6 Lectures)

Polymer processing- Melt: Thermoplastics and Thermoset: Processing thermoplastics- rotational moulding, injection moulding, compression moulding; melt spinning; processing thermosets- hand lay-up process, spray lay-up method, vacuum bagging, resin transfer moulding, resin infusion process; dry spinning, Spun bonding. (8 Lectures)

Polymer processing- Solution: Thermoplastics: Electrospinning; centrifugal spinning; wet spinning. (3 Lectures)

Composites: Importance of composites; classification- fiber and particle reinforced; factors affecting composite performance; failure. (3 Lectures)

Case Studies: Application of polymers in special cases-vibration damping; energy; adhesives; bio-medical-drug delivery, tissue engineering; water and air purification; electronics. (4 Lectures)

Recycling of Plastics and Environmental Issues: Need of recycling; degradation- thermal, chemical, hydrolysis, mechanical; incineration; bioplastics; issues with recycling. (3 Lectures)

Text Books

1. Robert O. Ebewele, **Polymer Science and Technology**, CRC press, 2000.
2. Richard A. Pethrick, **Polymer Science and Technology for Scientists and Engineers**, Whittles Publishing, 2010.